
DL for ENG
Designed by Remus Teodorescu 2026

Who Is Teaching This

REMUS TEODORESCU

- **Professor, AAU Energy** — power electronics, batteries, and control for the grid
- **thirty years of physics-based models** — and five of asking what data adds to them
- **this course** — twelve lectures written, and rewritten, over the past year
- **with Noman Khan** — research assistant, who runs the exercise sessions with you and has reviewed every notebook
- **the honest part** — the material was developed with AI assistance. I will say so wherever it matters.

WHY THIS COURSE EXISTS

the problem	energy systems have physics AND data
the usual answer	pick one and ignore the other
this course	put the physics inside the learning
the name for it	physics-informed deep learning

A FIRST EDITION

Everything here is new this year. You are its first readers, and its reviewers.

ret@et.aau.dk · nomank@energy.aau.dk

The Course in One Picture

ONE METHOD, TWELVE LECTURES

- **Part 1 · L1 – L6** — what a network is, how it learns, and why it works
- **the join · L6.1** — a loss term need contain no data at all
- **Part 2 · L7 – L12** — the physics goes into the loss
- **the applications** — heat, fluid flow, batteries and fuel cells, a robot car, the power grid
- **L13** — your mini-project, on your own topic

WHERE YOUR PROJECT MIGHT LAND

thermal	L8 — heat transfer, steady and transient
flow	L9 — Burgers, channel flow
electrochemical	L10 — Li-ion cells, solid oxide cells
autonomous	L11 — camera navigation on a JetRacer
grid	L12 — state estimation, stability

If your project is none of these, that is not a problem. It is the sixth row, and mini-projects are chosen at L13.

Who Is in the Room

THIRTY SECONDS EACH

- **your name** — and the programme you are on
- **your background** — mechanical, electrical, energy, something else
- **your semester project** — in one sentence
- **what it is trying to do** — predict, control, design, or understand
- **your Part 2 topic** — heat, flow, electrochemistry, robots, or the grid: the one closest to your programme or project. Decided today; it fixes which Part 2 exercise you hand in.

SEVEN PROGRAMMES, FIVE PART 2 TOPICS

PED · EPSH	Aalborg · power grid stability · Ex12
MCE · AIAS	Aalborg, Esbjerg · robot navigation · Ex11
HYTEC	Aalborg · electrochemical devices · Ex10
OES	Esbjerg · fluid flow · Ex09
APEL	Esbjerg · heat transfer · Ex08, or by project

A default by programme; your semester project can override it. The exercises cross all five, so different backgrounds are an asset here.

THE QUESTION FOR THE ROOM

In one sentence: what is your semester project trying to predict, control, or design — and which of the five Part 2 topics is closest to it?

Eight Weeks, Thirteen Lectures

PART 1

L1 · L2	Python, NumPy, SciPy, PyTorch 2 – 9 Sep · at home · recorded
L3	What is AI, and what it has built Wed 9 Sep · 12:30 – 16:15 · today
L4	Perceptron to deep networks Wed 16 Sep · 12:30 – 16:15
L5	CNNs, GNNs, sequences, attention Mon 21 Sep · 12:30 – 16:15
L6	Loss, gradients, post-training Fri 25 Sep · 10:15 – 15:15

PART 2

L7	PINN fundamentals, the PDEs Wed 30 Sep · 12:30 – 16:15
L8	Heat transfer, static and dynamic Fri 2 Oct · 10:15 – 14:15
L9	Laminar and turbulent flow Wed 7 Oct · 12:30 – 15:15
L10	Electrochemical devices, SOC Wed 14 Oct · 12:30 – 16:15
L11	JetRacer: navigation, driving Fri 16 Oct · 10:15 – 14:15 · ESBJERG
L12	Power grid stability Mon 26 Oct · 12:30 – 16:15
L13	Mini-projects Thu 29 Oct · 8:15 – 12:00

All sessions in Aalborg except L11, the JetRacer day in Esbjerg. Each lecture is two sublectures with a break; L3.2 is recorded, like L1 and L2.

Live Lectures and Recorded Ones

TWO TRACKS, SAME MATERIAL

- **recorded only** — L1, L2, and L3.2: watch at your own pace, before the lecture that needs them
- **live** — everything else, in this room, with a recorded version published afterwards for review
- **pause points** — each recording stops twice and asks you to work something out before it continues
- **the recordings are narrated by a synthetic voice from my script** — say if that gets in the way

HOW TO USE A RECORDING

before	watch, and stop at the pause points
during	have the notebook open beside it
after	replay the ten questions at the end
do not	watch at double speed the night before

THE QUESTION FOR THE ROOM

Did you watch L1 and L2 — all, some, skimmed, or not yet? And what would have made you watch more?

The Exercises Run in Your Browser

GOOGLE COLAB, NOTHING TO INSTALL

- **github.com/RemusTeodorescu/DL-for-Engineers-Public-Course** — every lecture PDF and every exercise notebook, always the latest version
- **click the badge** — Open in Colab, at the top of each notebook
- **run the first cell** — it fetches the set's library files by itself
- **first, save a copy in Drive** — or your work vanishes when the tab closes
- **a Google account is all you need** — no Python installed, no GPU of your own

THE ONE HABIT THAT SAVES EVENINGS

File	Save a copy in Drive — before anything else
then	every edit auto-saves
the badge	always reopens the pristine original
lost work	almost always an unsaved tab

THE QUESTION FOR THE ROOM

Who has opened a notebook in Colab already? Did the first cell run without asking for anything?

Exercise, Report, Moodle — Within a Week

THE RHYTHM

- **each lecture has an exercise set** — notebooks in numbered order, ending in a report with your measured numbers
- **Part 1, Ex01 – Ex06** — mandatory for everyone: six reports, one per set
- **Part 2, one set mandatory** — the topic from the intro round, your seventh report; the other sets are open, not handed in
- **hand in on Moodle within one week** — the reports are the course work, and the oral exam builds on them
- **late** — say so on Moodle before the deadline, and we agree a date

YOUR SEVEN REPORTS

Ex01 – Ex06	everyone · Part 1 · six reports
+ one of Ex08 – Ex12	heat · flow · electrochemical · robots · grid, the topic from the intro round
numbers	yours, not the slide's
a judgement	which model, and why — defended
a check	what you tried that did not work
the seeds	if a number depends on chance, say so

THE QUESTION FOR THE ROOM

Ex01 and Ex02 — how far did you get, and which cell cost the most time: the Python, the maths, or the notebook environment?

The Mini-Project

YOUR TOPIC, THIS METHOD

- **chosen at L13** — 29 October, after all twelve lectures
- **close to your semester project** — the point is to use this on something you already care about
- **alone or in a group** — your choice; a group shares the modelling, not the writing
- **the shape** — a physics-informed model of one thing from your project, with a measured comparison
- **support** — Noman in every exercise session, the L13 session, and the Moodle forum until the hand-in

WHAT COUNTS AS A RESULT

a baseline	the physics-only or data-only answer
the model	physics in the loss, trained by you
the comparison	measured, with its uncertainty
the judgement	would you use it? Under what conditions?

THE QUESTION FOR THE ROOM

Looking at the board from the intro round: which of your projects could become a mini-project, and which two might share one?

Where Things Happen

ONE PLACE FOR EVERYTHING

- **Moodle** — announcements, hand-ins, the forum, links to the videos
- **the public repository** — slides and notebooks, always the latest version
- **the Moodle forum before email** — a question one of you has, ten of you have; Noman and I both read it
- **I read the forum before each lecture** — and answer what needs answering there, for everyone at once
- **exercise sessions** — Noman Khan is there for every one; office hours with me by appointment through Moodle

THE FEEDBACK DEAL

you find	a cell that fails, a claim that is wrong, a slide that confuses
you post	on the Moodle forum, with the notebook and cell
I fix	usually the same week, and say so
you see	the fix on the badge, immediately

THE QUESTION FOR THE ROOM

What is one thing you would like different by next week — in the videos, the notebooks, or the way the sessions run?

You Will Use a Language Model. Everyone Does.

THE ONLY QUESTION IS HOW

- **paste a TODO cell into any model** — it will write the answer in seconds, and it will usually be right
- **so the assignment is not the point** — the assignment is the occasion
- **what I assess** — whether you can explain, defend, and adapt what you hand in
- **what an engineer is paid for** — the judgement, not the typing
- **the rest of this part** — what the difference looks like, and the rule for this course

TWO STUDENTS, SAME SUBMISSION

student A	pasted, ran, submitted. Twenty minutes.
student B	asked, read, changed a line, asked why. Two hours.
in November	identical reports
in March	only one can build the next thing

THE QUESTION FOR THE ROOM

Hands up: who used a language model on Ex01 or Ex02? Nobody in this room will be marked down for that answer.

From General Questions to Your Engineering Problem

YOU ALREADY USE THESE TOOLS — JUST NOT FOR THIS

- **how most people use them** — summarise, translate, draft, explain a concept, fix a line of code
- **what an engineering problem needs** — a specific system, units, an equation, a measurement to check against, a model somebody must trust
- **the gap** — a general answer is not an engineering answer. It has no plant, no sensor, no failure mode.
- **what this course teaches** — how to take a tool that answers general questions into your own specific project, step by step
- **the progression** — a Python tutor in L1 – L2, a reasoning partner in Part 1, a collaborator on your physics in Part 2, a colleague on your mini-project

THE SHIFT, IN FOUR MOVES

from a topic	to your system, with its numbers
from an answer	to a prediction you then measure
from trusting it	to checking it against the physics
from asking	to specifying, and defending the result

THE QUESTION FOR THE ROOM

What was the last thing you asked a language model — and was it anything like an engineering problem?

The Rule for This Course

ALLOWED, FOR EVERYTHING

Any tool, any time: writing code, debugging, explaining a slide, drafting a report. This is how engineering is done now, and pretending otherwise would teach you a lie.

DECLARED, IN THE REPORT

One line: which tool, for what. Not a confession — a method statement, like naming the solver you used.

OWNED, LINE BY LINE

Anything you hand in, you can explain when asked. If you cannot, it is not yours yet — go back and make it yours.

REPORTS, EXAM, MINI-PROJECT

written reports	always state which tool you used, and for what
exam · preparation	everything allowed, any tool
exam · the 20 minutes	nothing allowed — you, the board, and the question
in the mini-project	a live discussion of your model
never	a tool detector, or a trap

The twenty minutes are the whole point of the rule above: what you own, you can explain without a tool.

What Deeper Learning Looks Like

THREE THINGS A PASTE CANNOT GIVE YOU

- **prediction** — say what a change will do before you run it. Right or wrong, you learn either way.
- **failure** — the exercises include runs that go wrong on purpose. The report asks what you saw.
- **transfer** — the same idea in a new place. Weight sharing appears three times in two lectures; noticing is the learning.
- **the model can do all three with you** — ask it to quiz you, not to answer you

PROMPTS THAT TEACH

"before I run this"	"what should the loss do, and why?"
"I got X"	"what are three reasons it is not Y?"
"explain the line"	"then ask me to explain it back"
"disagree with me"	"where is my reasoning weakest?"

THE QUESTION FOR THE ROOM

Which of those four prompts have you ever used — and which would feel strange to try?

Where the Energy Sector Is Hiring

THE INTERSECTION

- **most AI courses teach you to fit data** — most energy courses teach you the physics
- **this one sits at the intersection** — and that is where the sector is hiring
- **in engineering you usually know the equation** — discard it and fit a curve, and the model fails the moment conditions leave the training data
- **physics in the loss instead** — the model respects conservation, and can be trusted further from the data it saw
- **not a future skill** — the companies on the right are doing it now

IN PRODUCTION, IN DENMARK AND BEYOND

Energinet	AI forecast of reserve capacity on the Danish grid — about a million kroner a week saved
Ørsted	machine learning across 5.5 GW of wind, solar and storage
Vestas · Grundfos	fleet-wide blade data; machine-health models against unplanned downtime
Siemens Energy · NVIDIA	an industrial AI operating system across the life of physical plant

THE QUESTION FOR THE ROOM

You will graduate into an industry that expects physics and networks together. Which of these companies, or which other, do you have your eye on?

company reports and press releases, 2025 – 2026

Four Things You Will Be Able To Do

BUILD A MODEL FROM DATA, AND SAY WHAT IT ASSUMES

A trained network, and the sentence about its noise model, its capacity, and the seeds it was checked on. Most published results skip that sentence.

PUT AN EQUATION INSIDE THE LEARNING

A physics-informed network for a heat, flow, cell, or grid problem — trained with no measurements at all, then with some.

DEPLOY ONE, AND MEASURE WHAT IT COSTS

Quantised, timed, on a device that also drives the motors. Latency budget in metres, not milliseconds.

AND THE FOURTH

judge	when not to use any of it
because	a solver that runs in a second wins
because	thin data deserves a curve, not a network
because	certification wants bounds, not error bars

The fourth is the one that makes the other three worth having. It is also the one interviewers ask about.

Back to Your Projects

THE LIST FROM THE INTRO ROUND

- **for each project on the board** — which of the four capabilities would move it forward?
- **the honest answer for some** — none yet, or not this semester. That is fine.
- **the mini-project is where this closes** — your topic, this method, measured
- **and the semester project is the reason to care** — not the grade

WHAT TO DO THIS WEEK

today	L3.1 after the break; L3.2 on video
this week	Ex03 — one notebook, your first model
by Wed 16 Sep	the Ex03 report on Moodle
if L1 – L2 are unwatched	watch them before Ex03. It assumes them.

THE QUESTION FOR THE ROOM

Look at your own project on the board: which one of the four capabilities is closest to it?

BREAK

Fifteen Minutes. Then Lecture 3.1

What artificial intelligence is, how it got here, and where an engineer should and should not reach for it. No mathematics beyond one energy function.

WHAT I TAKE FROM THIS SESSION

the board	your projects, under the five application rows
the videos	what stopped you watching, if anything
the notebooks	which cell cost the most time
one thing	you want different by next week

All four go into the material this week. You will see the changes on the badge, and I will say what changed at the start of L4.

If Asked: What Ex03 Contains

YOUR FIRST MODEL, ONE NOTEBOOK

- **a vibrating mass** — a signal you can write the physics of, and measure
- **fit a network to it** — and watch it fail to extrapolate
- **then add the physics** — and watch the same network extrapolate
- **the report** — the extrapolation plot, and one paragraph on why
- **the point of the whole course** — in one notebook, before the theory

IT ASSUMES

from L1	functions, arrays, shapes
from L2	a tensor, and .backward()
from nobody	any machine learning
time	two to three hours

If Asked: How the Recordings Were Made

HONESTLY

- **the slides** — generated from code, one script per lecture, so every deck agrees on style
- **the script** — written by me for a listener, with two pause points per lecture
- **the voice** — synthetic, from that script. It is not me. It says what I would say.
- **the cursor** — highlights the part of the slide the script is pointing at
- **the reason** — twelve lectures, each in a live and a recorded form, in one year

WHAT AI DID, AND DID NOT

did	draft, check, rewrite, render, narrate
did not	decide what the course teaches
did not	grade anything, ever
will	get better when you tell me what is wrong

If Asked: The JetRacer Day

FRIDAY 16 OCTOBER, ESBJERG

- **six Waveshare JetRacer Pro cars** — a camera, a Jetson Nano, four 18650 cells
- **you drive, it records** — then a network learns to steer from your driving
- **then it drives** — and you measure how far it travels blind between two decisions
- **groups of three or four** — one car each, all morning
- **travel** — plan the trip to Esbjerg; details on Moodle by L9

BEFORE THAT DAY

L5.1	convolutional networks — what the car sees
L6.2	deployment — why the car cannot wait
Ex11 notebook 00	run it the week before
a charged laptop	and closed-toe shoes

BACKUP

If Asked: Reading

THREE BOOKS, NOTHING TO BUY

- **Prince** — Understanding Deep Learning, MIT Press 2023. The main text for Part 1, free on the author's site.
- **Goodfellow, Bengio, Courville** — Deep Learning, MIT Press 2016. For the optimisation chapters, free online.
- **G. R. Liu** — PINN with Python: An Introduction. A fifty-page excerpt on Moodle, the one text for Part 2.
- **each lecture names its chapter** — usually twenty pages, read before or after
- **in Part 2 the papers are cited on the slides** — the excerpt and the exercises are the text

FOR THIS WEEK

read	Prince chapter 1 — twenty minutes
watch	L3.2 — what the field has built
run	Ex03
bring to L4	one model from your own work